LIAM ISAACS

Principal Engineer

liamnisaacs@gmail.com | (650) 208-4292 | github.com/yi-ye-zhi-qiu | liamisaacs.dev

5+ years building large-scale web data systems (1B+ signals/day, 60M+ requests/day). Founded and led data engineering and frontend teams, architecting end-to-end infrastructure from crawling to customer-facing product.

EXPERIENCE

Daash — daash.co New York, NY | May 2021 – Present

Principal Engineer, Founding Employee

Web Crawling Infrastructure

Founded and led Data Acquisition team (5 engineers)

- Built distributed crawling system processing 1B+ web signals/day with 98% uptime over 4 years
- Wrote 50+ high-performance Scrapy crawlers (50k+ pages/min) with bot-evasion capabilities, including a broad-crawl system spanning 10k+ domains
- Architected serverless AWS infrastructure (Terraform, Docker, GitHub Actions CI/CD) with 1–3 minute deployments per crawler
- Designed custom ML pipeline (agglomerative clustering + KMeans) reducing scrape volume by 80%, cutting costs while preserving informational value
- Developed statistical sampling methodology saving up to 90% on scrape volume, strengthening crawler defensibility

Data Platform & Analytics

Founded and managed data engineering team (10 engineers)

- Architected bronze/silver/gold medallion ETL pipeline (Python, SQL, Athena/S3) in a GitHub monorepo, normalizing unstructured web data through precompute layers with full PyTest coverage
- Achieved 10–15 minute end-to-end pipeline loads with transparent, auditable data management (write-audit-publish framework)
- Built time-series models (Python pandas, operationalized in SQL) generating production sales estimates, validated against ground truth via MAE and Pearson correlation

Frontend & Product

Led frontend team (2 engineers)

- Independently architected and shipped production analytics dashboard in Next.js 15 app router design / React 19 (TypeScript, Tailwind, ShadCN, TanStack Query, Zustand) with DAL/DTO architecture (plus API routes), deployed on Vercel
- Implemented full CRUD (DynamoDB), federated data queries (Athena), and user auth (AWS Cognito), achieving sub-2s latency per user query

PROJECTS

leeuhm.xyz — *source*

- Full-stack Next.js app (app router) serving SHAP visualizations of a logistic regression model trained on scraped League of Legends data; built in 24 hours as part of a live technical lecture

liamscafe.com — *source*

- Personal site built with Next.js, Tailwind, and 3D models (Blender / React Three Fiber)

EDUCATION

Wesleyan University — B.A. Chemistry (3-year graduate) 2016 – 2019

- Princeton in Beijing Full Scholarship (2018)
- Elizabeth Verveer Tishler Piano Competition Finalist (2018, 2019)

LANGUAGES & INTERESTS

Languages: English (native), Mandarin (partial fluency), Ukrainian (A2)
Interests: Piano, badminton, rock climbing, running, cats